Calibration of Measurement Equipment

Purpose

This work instruction defines the procedure for calibration and verification of measurement and test equipment at Acme Corporation.

Scope

This instruction applies to all measurement and test equipment used for product acceptance, process control, and quality verification.

Definitions

Calibration: Comparison of a measurement device against a known standard to detect and adjust any deviation.

Traceability: The property of a measurement result whereby it can be related to a national or international standard through an unbroken chain of comparisons.

Due Date: The date by which the next calibration must be completed.

Procedure

Step 1: Equipment Identification

Each measurement device is assigned a unique equipment ID and recorded in the calibration management system. The equipment record includes:

- Equipment ID and description
- Manufacturer and model number
- Serial number
- Range and resolution
- Accuracy specification
- Calibration procedure reference
- Calibration frequency
- Location

Step 2: Calibration Scheduling

The calibration management system generates a calibration schedule. Equipment is calibrated at the following intervals:

Equipment Type	Calibration Interval
Pressure gauges	12 months
Calipers and micrometers	12 months
Torque wrenches	6 months
Electronic test equipment	12 months

Equipment Type	Calibration Interval
Temperature probes	6 months
Weighing scales	12 months
Go/No-go gauges	24 months
Hardness testers	12 months

Step 3: Calibration Execution

Before calibration:

1. Verify the calibration standard has valid traceability to national standards.
2. Clean the equipment and inspect for damage.
3. Allow the equipment to stabilize at ambient temperature for 2 hours.

During calibration:

1. Perform calibration at minimum 5 points across the measurement range.
2. Record all readings on the calibration data sheet.
3. Apply environmental corrections if required.
4. Determine if the equipment is in tolerance.

After calibration:

1. If in tolerance, apply a calibration sticker with the due date.
2. If out of tolerance, tag the equipment as "OUT OF CALIBRATION" and initiate an NCR per SOP-QA-012.
3. Update the calibration management system.

Step 4: Out of Tolerance Response

When equipment is found out of tolerance:

1. Determine the last calibration date when the equipment was in tolerance.
2. Identify all products measured with the equipment since that date.
3. Evaluate the impact on product quality.
4. Recall and re-inspect affected products if necessary.
5. Document the assessment in the NCR.

Step 5: Records

Calibration records are retained for the life of the equipment plus 5 years. Each record includes:

- Equipment ID and description
- Calibration date and due date
- Calibration results and as-found/as-left data
- Calibration standard used (with traceability)
- Technician name and signature
- Approval by QA Supervisor

Related Documents

- SOP-QA-012: Non-Conformance Report Procedure
- WI-034: Production Line Pressure Monitoring

Document Version 1.3 — Last Updated March 2026